

Project Title: Economic Analysis of Autonomous Site Monitoring Systems for Large-Scale Construction Projects

Team Members:

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Brief Project Definition

This project evaluates the feasibility of implementing an automated, AI-driven monitoring system for a high-rise construction site. Traditional site monitoring relies on manual inspections and human supervisors to track progress and ensure safety compliance. This investment proposes replacing or augmenting manual supervision with a network of IoT sensors, edge-computing cameras, and automated reporting software.

From a Construction Engineering perspective, we will analyze the impact on project timelines and labor costs. From a Computer Engineering perspective, we will analyze the hardware lifecycle, data processing infrastructure, and system maintenance. The goal is to determine if the reduction in human error and accident-related costs justifies the high initial investment in technology.

Preliminary Listing of Alternatives

We will examine the following three alternatives to determine the most cost-effective approach over a 5-year period:

- **Alternative 1: The "Manual" Baseline (Status Quo)** Continuing with traditional human-led safety and progress monitoring. This involves high recurring labor costs (salaries, insurance, benefits) but zero initial capital investment in specialized monitoring technology.
- **Alternative 2: Leasing a SaaS Solution** Renting a complete system (Hardware + Software) from a third-party provider. This converts the investment into a monthly operating expenditure, reducing initial risk but resulting in higher long-term costs without asset ownership.
- **Alternative 3: In-House Development and Ownership (Capital Investment)** Investing in the purchase of the hardware (cameras, sensors, servers) and the development of custom monitoring software. This requires a large initial capital expenditure and annual

maintenance costs but provides the team with a depreciable asset and lower costs in the long run.